

AI-Powered Fleet Performance & Delivery Efficiency Dashboard

Objective

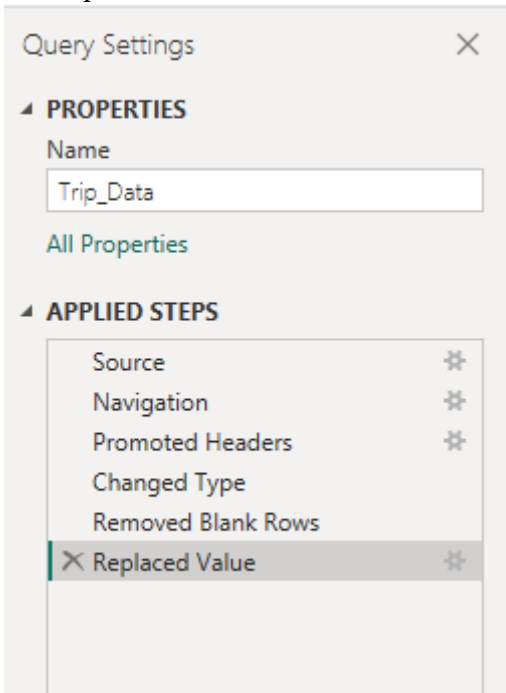
A logistics company wants to analyze fleet performance, delivery efficiency, fuel consumption, and operational costs. The dashboard enables stakeholders to make data-driven decisions by monitoring key metrics and identifying inefficiencies.

Data Overview

The dataset includes information related to trips, vehicles, drivers, destinations, fuel consumption, maintenance cost, distance traveled, delivery status, and time-based attributes.

1. Data Cleaning & Modeling:

- Removed duplicate trip records



- Late delivery fuel values were imputed using average fuel consumption for late deliveries

```
= Table.ReplaceValue("#Removed Blank Rows",null,91.57,Replacer.ReplaceValue,{"Fuel_Consumed_L"})
```

Replace Values

Replace one value with another in the selected columns.

Value To Find

null

Replace With

91.57

OK

Cancel

- Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.

Properties >>

Relationship

Table

Trip_Data

Column

Vehicle_ID

Cardinality

Many to one (*:1)

Table

Vehicle_Master

Column

Vehicle_ID

Make this relationship active

☒ Yes

Cross-filter direction

Single

Apply security filter in both directions

☐ No

Apply changes

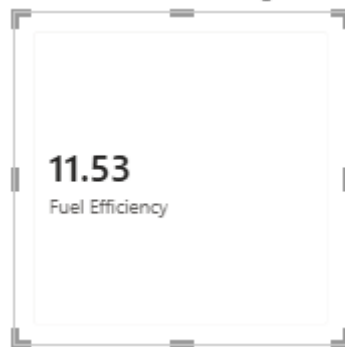
Open relationship editor

2. DAX Measures:

Created measures for analysis

- Fuel Efficiency = Distance / Fuel Consumed

```
1 Fuel Efficiency = DIVIDE(sum(Trip_Data[Distance_km]),sum(Trip_Data[Fuel_Consumed_L]))
```

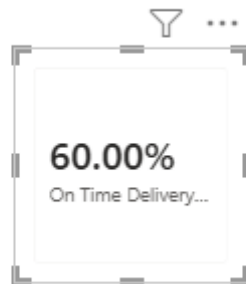


- On-Time Delivery % = On-Time Trips / Total Trips:

```
1 On Time Trips = CALCULATE(COUNT(Trip_Data[Trip_ID]),Trip_Data[Delivery_Status]="On-Time")
```

```
1 Total Trips = COUNT(Trip_Data[Trip_ID])
```

```
1 On Time Delivery % = DIVIDE([On Time Trips],[Total Trips])
```



- Cost per km = (fuel cost + Maintenance Cost) / Distance

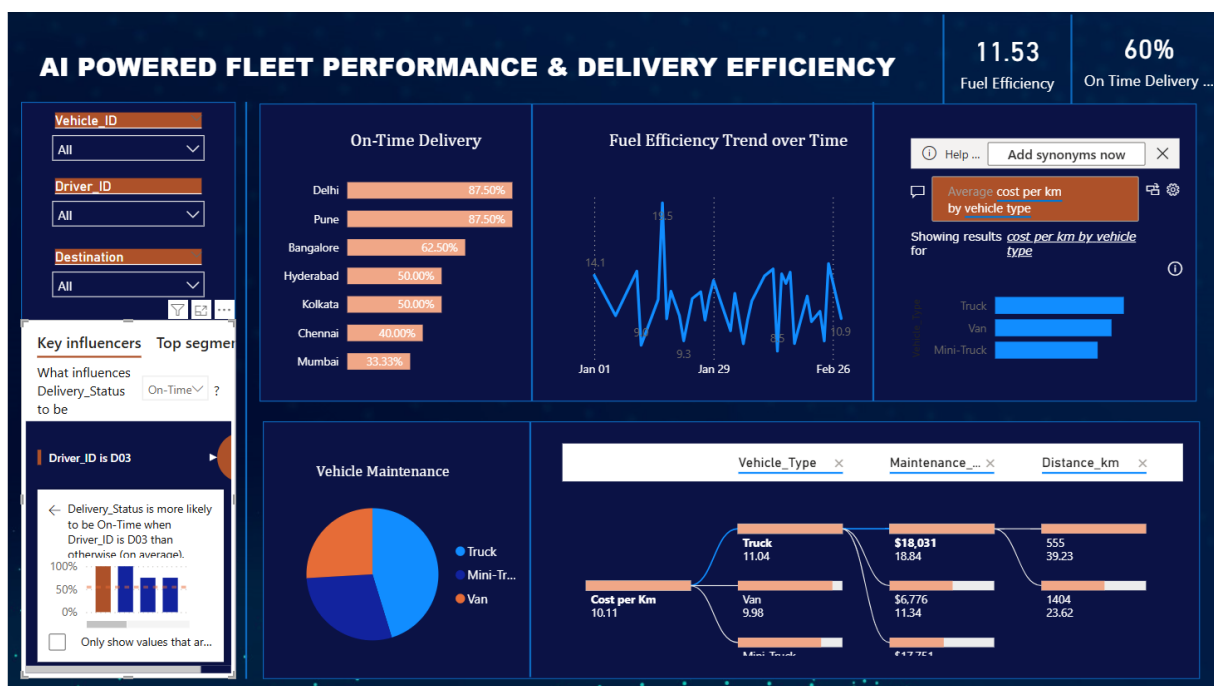
Note fuel cost =100

```
1 Maintenance Cost = SUM(Vehicle_Master[Maintenance_Cost])
```

```
1 Cost per Km = DIVIDE([Fuel cost]+[Maintenance Cost],SUM(Trip_Data[Distance_km]))
```

3. Dashboard Visuals Explanation

- Vehicle Type slicer for fleet comparison
- Driver ID slicer for driver performance
- Destination slicer for location-wise analysis
- KPI Cards: Display Cost per Km, Fuel Efficiency, and On-Time Delivery Percentage
- Bar Chart: On-Time Delivery Percentage by Destination
- Line Chart: Fuel Efficiency trend over time
- Pie Chart: Maintenance cost distribution by vehicle type
- Key Influencers Visual: Identifies factors influencing delivery delays
- Decomposition Tree: Breaks down Cost per Km by Vehicle Type, Maintenance Cost, and Distance



Conclusion

This dashboard provides a comprehensive view of fleet operations and delivery performance. It helps identify cost drivers, optimize vehicle usage, and improve delivery reliability.